Contest Syllabus

Grade 3–4 Contest Syllabus

Section 1: Introduction to AI

1. **Understanding Intelligence**:
   - What makes humans intelligent?
   - How is machine intelligence similar yet different?
2. **Definition of AI**:
   - How AI make smart machines that learn and make decisions

Section 2: AI in Daily Life

1. **AI-Powered Devices**:
   - Examples: Voice assistants (Alexa, Google Home).
   - How they recognize and respond to human commands.
2. **AI in Entertainment**:
   - AI behind recommendations (e.g., YouTube, Netflix).
   - AI in gaming: How games "learn" to challenge players.

Section 3: Comparing Humans and Machines

1. **Cognitive Abilities vs. Computational Power**:
   - Humans excel in creativity; machines excel in repetitive tasks.
   - Examples: Chess-playing AI vs. storytelling by humans.

Section 4: AI Learning Basics

1. **How AI Learns**:
    - Simple explanation of data feeding and pattern recognition.
2. **Introduction to Machine Learning**:
    - AI learning from labeled data to identify objects (e.g., photos of cats vs. dogs).

Section 5: Creative and Useful

1. **AI for Creativity**:
    - How AI generates paintings, music, and stories.
    - AI-based apps for drawing and coloring.
2. **AI in Problem Solving**:
    - AI in navigation apps (Google Maps) and translating languages.

Section 6: Ethics in AI

1. **Good and Harmful Uses of AI**:
    - Protecting nature with AI-powered drones.
    - Risks of AI in spreading fake news.
2. **Responsibility in Using AI**:
    - Teaching kids to use AI ethically (e.g., not cheating in games).

Grade 5–6 Contest Syllabus

Section 1: Understanding AI

1. **What is AI?**
   - Explanation and importance.
   - How it is different from regular computer programs.
2. **Types of AI**:
   - Narrow AI vs. General AI.

Section 2: AI Learning Process

1. **How AI Learns**:
   - Learning from data (e.g., recognizing shapes or objects).
   - Examples of AI learning in games and quizzes.

Section 3: Applications of AI

1. **AI in Daily Life**:
   - Translators, calculators, search engines
2. **AI in Industries**:
   - Healthcare: Disease detection.
   - Agriculture: Monitoring crops.
   - Education: Personalized learning.

Section 4: Creativity with AI

1. **AI Generating Art and Music**.
2. **Exploring AI Tools in Creativity**.

Section 5: Ethics and Future of AI

1. **Basic AI Ethics**:
   - Fairness and responsibility.
2. **What Can AI Do in the Future?**

Grade 7–8 Contest Syllabus

Section 1: Machine Learning Basics

1. **What is Machine Learning?**
   - How AI learns from data.
2. **Components of ML**:
   - Data, algorithms, and models.

Section 2: Types of Machine Learning

1. **Supervised Learning**:
   - Learning from labeled data.
2. **Unsupervised Learning**:
   - Finding patterns in unlabeled data.

Section 3: AI and ML in Action

1. **Applications of ML**:
   - Voice assistants.
   - Search engines.
   - Recommendations (e.g., Netflix, YouTube).

Section 4: Real-World Problem Solving with ML

1. **Solving Everyday Problems**:
    - AI in navigation apps.
    - AI in smart homes.

Section 5: Ethics in ML

1. **Understanding Bias in AI**:
    - What happens when AI is unfair?

I Am SCIENTIST

Grade 9–10 Contest Syllabus

Section 1: Advanced AI and ML Concepts

1. **What is AI?**
   - Its purpose and goals.
2. **Types of AI**:
   - Narrow AI, General AI and Superintelligence.
3. **Machine Learning Basics**:
   - How machines improve through experience.

Section 2: AI Algorithms and Models

1. **Introduction to Algorithms in AI**:
   - Regression and classification.
   - Decision-making processes.
2. **Data in AI**:
   - Collection, preprocessing and quality assurance.

Section 3: AI Applications

1. **AI in Various Fields**:
   - Finance: Fraud detection.
   - Healthcare: Diagnostics.
   - Entertainment: Recommendations.

2. **AI in Robotics**:

    - Navigation and automation.

Section 4: Ethics and Impact of AI

1. **Ethics in AI**:

    - Algorithmic bias.
    - Privacy concerns.

2. **Implications of AI on Jobs and Society**.

Grade 11–12 Contest Syllabus

Section 1: Advanced AI Concepts

1. **Deep Learning Basics**:
   - Structure of neural networks.
   - Difference from traditional ML.
2. **Convolutional Neural Networks (CNNs)**:
   - Image recognition and processing.
3. **Recurrent Neural Networks (RNNs)**:
   - Applications in language and sequence data.

Section 2: Data Science and Big Data

1. **Role of Data in AI**:
   - Handling large datasets.
   - Tools for data analytics.

Section 3: Applications of Advanced

1. **Natural Language Processing (NLP)**:
   - Sentiment analysis.
   - Chatbots and translation.
2. **AI in Cybersecurity**:
   - Threat detection.
   - Malware analysis.

Section 4: Cutting-Edge AI

1. **Reinforcement Learning**:
    - Concepts and real-world applications.
2. **AI in Quantum Computing**:
    - Basics and potential impact.
3. **AI in Cloud Computing**:
    - Enhancing AI capabilities.

Section 5: Ethical and Social Implications of AI

1. **Ethics in Deep Learning**:
    - Challenges with transparency.
2. **Bias in Neural Networks**:
    - Ensuring fairness in AI systems.